

Queering the Space: Location-Based Stories for Transforming Space into Place

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ABSTRACT

Being part of a minority group can be isolating and lonely; in some places, members of LGBTQ+ groups build up communities and neighborhoods, but in other areas, LGBTQ+ places are more invisible. In this paper we present a system that facilitates a digital layer of stories and experiences by other LGBTQ+ individuals. We present our novel prototype and our user study of the application, where users discuss how this digital service can help building place based community among LGBTQ+. This study contributes by providing a better understanding of how the physical space of the city can be transformed into meaningful places through a digital layer of personal stories to encourage connections within the LGBTQ+ population.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; *Empirical studies in collaborative and social computing*.

KEYWORDS

Location-Based Services, Place, Interactive Experiences, Mobile Applications, LGBTQ+

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1 INTRODUCTION

The notion that digital technologies can contribute to a more meaningful urban community has been discussed within both HCI and CSCW research [7, 13, 24]. Particularly the reasoning of how spaces become places, either digitally represented as virtual worlds, or as digital layers on top of geographical places, has been predominant. While space is the “structure of the world”, that is the environment in which we move around [13], place possesses value, understood behavioral appropriateness and cultural expectations [ibid]. While most sub-communities use an array of physical signs to signal

presence and present place as such, it is not always enough for minorities to feel a close affiliation, or even notice their community’s presence. While digital technologies can sometimes help create a layer of awareness of particularly places of interests (e.g. digital maps having descriptions of restaurants and bars as ‘gay bar’, or a church of a specific denomination), these digital maps do not describe people’s felt experiences as such. We suggest that there are several missed opportunities for adding a more personal, meaningful layer to physical space, and creating places with more meaning and community.

In this paper, we present a location-based application we designed and developed in order to investigate what impact queer stories might have on LGBTQ+ individuals and their understanding of public spaces. Specifically, we explored how this group of people experience spaces and places in Copenhagen, Denmark, and how these perceptions can be impacted by a digital layer of stories. We engaged with participants through interviews to gain insights into their perceptions and understandings of the concept. Our study contributes to existing research on location-based technologies and their relevance to LGBTQ+ people in creating places with meaning to this particular group of people.

Note that our study was of places in a very LGBTQ+ friendly city, in a country that prides itself in being the first to legalize same-sex unions in 1989; while the system and study are rooted in a liberal context, Copenhagen is still not without occasional discrimination and lack of inclusivity of LGBTQ+ identifying people. The system would be difficult and in some contexts even dangerous to implement in other countries where these populations are not accepted, and in some cases are even illegal. As such, this research should be viewed in the context of a liberal, LGBTQ+ friendly society. In this paper we use the terms “queer” and “LGBTQ+” interchangeably, as “queer” in some contexts can refer to people identifying as non-heterosexual and/or non-cisgender individuals.

2 RELATED WORK

Our related work spans a broad area of location based services, the understanding of space and place, as well as LGBTQ+ spaces and identity. We limit our scope here by focusing on the scholarly discussion of space and place within HCI, location based experiences relevant for our own application and study, as well as the presence of LGBTQ+ communities within urban settings.

2.1 Space and Place Technologies

The notion of digital place and the relation between space and place have both been part of a long-running discourse within HCI. One of the earliest contributions to understanding the distinction between place and space (in relation to digital technologies) was the 1996

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paper by Harrison and Dourish, in which the authors make a distinction between place and space, for the purpose of understanding design of communication technologies, for example virtual places that have social meaning, but not necessarily a physical space to be located in. While physical space frames our behavior, our everyday experience and interaction gives us a sense of place [13]. Spurred by the development of location tracking technologies, research involving or focusing on location in relation to place making has increased significantly, such as location sharing using places rather than map coordinates [1, 21], place making for urban commuting [17] and focus on how people can incorporate playful aspects of location into social media [2, 15], just to name a few examples.

2.2 Location Based Experiences

A wide variety of studies have explored location-based technologies and how they assist people in everyday life, particularly as the emergence of the smart phone enabled location tracking and third-party applications. Some of the early work on location-based technologies was done by researchers and media artists who experimented with mobile devices focusing on how to locate users and things in physical space. They developed and analysed location-based mobile games, art projects and locative audio walks, emphasising how mobile devices can enhance the connection to the surrounding space [9]. In one of the early studies, researchers explored a location-based gaming concept using a mobile phone. The research concluded that real-life meaning and memories attached to locations influence the way the game is played, and users also gained more interest in going to unknown parts of their city [22]. Just a few years later, a study of the mobile game Feeding Yoshi showed similar findings: that people were able to integrate places into their play and went out of their way to find location specific game areas [3]. Another study explored the experiences from an interactive, location-based drama, and concluded that designing for loose coupling between the story and city scape can enable people's engagement in meaningful experiences [19].

Other researchers have investigated how to use the geolocation of mobile devices to retrieve and place digital information through various applications. One of these was GeoNotes, which attempted to democratize public space by allowing all citizens to add annotations to the city, regardless of commercial interests. GeoNotes invited users to place virtual post-it notes tied to specific locations for other users to find at the location where it has been placed [18]. Another relevant application and accompanying study was that of Columbus: a mobile application enabled the exploring of geotagged photographs at the location they were taken. The study pointed out that people in familiar environments appropriated the application to look for pictures according to their previous experience and knowledge of the place, whereas people unfamiliar with the environment were more explorative in their way of approaching the place [20].

2.3 Queer Spaces

While specific 'gay neighborhoods' developed between World War II and the late nineties, particularly in the US, these neighborhoods have recently experienced gentrification and out-phasing, due to increased "feelings of acceptance, integration, and safety" [10]. In

Europe, distinct LGBTQ+ neighborhoods were never as common, and in Copenhagen, where our study took place, there is no such distinct neighborhood, despite—or because of—the country's general liberal approach to the LGBTQ+ population.

Research within HCI concerning LGBTQ+ people has mostly focused on gay men and the use of dating and hookup apps [4, 5, 12]. A recent survey showed that while there has been a growth in research mentioning as well as focusing on LGBTQ+ people in HCI, most of the papers still only briefly mentioned them, rather than focusing exclusively on the population [23]. In regards to specific studies, one recent study looked at Grindr, a location-based social networking application that primarily targets gay men, in order to examine how such applications synthesize online queer encounters with offline physical spaces to create a new hybrid terrain. As such, they investigated questions about queer technologies and their impact on the queer community and a sense of place, concluding that such technologies have the potential to imbue physical spaces with a queer connection while at the same time decentralising the primacy of established gay neighbourhoods [16]. Finally, studies point out how well-being is affected by several dimensions of community for queer individuals, highlighting the importance of fostering dimensions of "being close, being read, being seen and being heard" [14].

3 METHOD

Our project focused on providing a digital layer of queer stories on top of places in a city, for encouraging community and connection. We first developed the application and collected stories, then studied how people would potentially use it and how this digital layer could benefit a sense of connection to their community.

3.1 The Queer Annotations Application

To explore the effect of a digital layer of stories, we developed "Queer Annotations", so named to indicate that stories are linked to other data such as physical location. These stories may include personal anecdotes, memories, or experiences associated with a specific location. The application was tailored specifically for the LGBTQ+ community, and was designed to be dependent on active users to generate content.

The application structure was simple: it allowed users to log in, create stories pinned to a specific location, and read or listen to stories when in a particular place. As a proof-of-concept application, we omitted security related and moderation features which would be necessary in a final product. Figure 1 shows a selected set of screenshots of the app.

3.2 Collection of Sample Stories

For our initial study, we decided to focus primarily on how location-linked stories uploaded by others impacted user experience of those locations. We thus pre-populated the app with a set of eight stores gathered via an online form shared across multiple LGBTQ+-oriented social media groups within the Copenhagen community. The form explained the app we were developing and noted what type of information would be used, the text is shown in table 1.

We requested four types of data for the stories: *username*, *title*, *description*, and *location*. We ensured stories were restricted to the

Table 1: Form text for collecting sample stories

Form text
This form is created to collect data for a user-generated queer platform, dependent on location-based technologies. The application will connect queer stories to locations in and around Copenhagen. By using the application users will be able to retrieve and read stories at the places they are connected to. This form will help us build early content for the platform which will be helpful for testing. For testing purposes the connected location should be the Inner City of Copenhagen within a distance of approximately 500m from Rundetårn. Your chosen name/username, title and story will be public—therefore you are more than welcome to use an alias or not disclose any name.

Table 2: Participants; names changed for anonymization

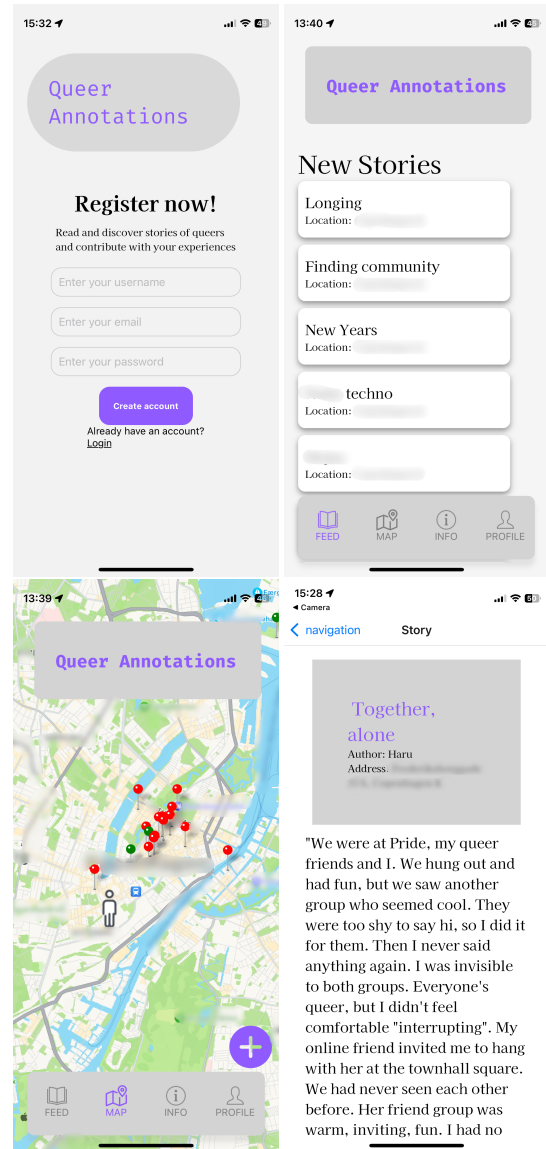
Name	Age	Gender	Occupation
Andy	23	cis male	Student (Rhetoric)
Jonathan	32	cis male	Project manager
Noah	29	non-binary	Language teacher
Sabrina	33	cis woman	Not provided
Simon	21	cis male	Student (Business)
Ulrich	35	cis male	Student (Urban Planning)

physical area of the user test, but in future use of the app, any location could be tagged.

3.3 Participants

For the study itself, we recruited six participants (Table 2). Two were recruited through the social media groups from which the collection of sample stories had also been gathered, and the remaining four via Grindr, a location-based social network for people identifying as gay, bisexual, trans and queer. A research profile on Grindr was created that explicitly stated the purpose of recruiting participants for tests and interviews. Some of the users were recruited through direct messages and others contacted the profile themselves. We do acknowledge that the majority of participants were gay men, and as such, the results have limited generalizability beyond that population.

The participants were met individually in the centre of Copenhagen where they received the application to begin the test, using their own phones. The initial tasks included signing them up as a new user, logging in and adding a story to the map, so they would get a feel for the functionality of the app. Then, we went for a walk around the city to open and read some of the stories at their associated locations. During the walk we inquired into topics such as their understanding of places, the significance of queer communities, and how they experienced the connection between the stories and the location. After each test, we conducted a semi-structured interview to ask about the participant's insights into the concept and how the application impacted their understanding of the places the stories

**Figure 1: Screenshots from the application: Registering, List of stories, Map of stories and sample story.**

were connected to. Testing and interviews were conducted by the first author.

We followed European guidelines for ethics in accordance to our universities' practices at the time of the study, including retaining informed consent and anonymization of data. A formal ethics process was not required at the time of the study.

3.4 Analysis

We collected observational notes from the walk and transcribed recordings of the interviews, and analyzed this data using thematic analysis [6]. After initial themes were suggested by the main researcher, two of the authors discussed the data and fine-tuned

themes. These themes were then saturated by interview and observational data.

4 RESULTS

Our aim with the study was to unfold people's experiences in regard to the impact the collected stories had on their understanding of the places. One of our hypotheses was that the application might change participant understanding of the place in question, similar findings in the literature on how location-based technologies can add a new digital layer to the understanding of a place [13]. We separate our findings into three parts: how users gained a sense of belonging to their community through the stories, how their sense of place was created or adjusted, and how they envisioned the application being beneficial to future connections.

4.1 Sense of Belonging and Community

As mentioned above, the stories presented in the application were gathered from various LGBTQ+ groups on social media, in an aim to make the content authentic and relevant to the participants, all of whom identified as LGBTQ+. Several of the participants expressed that they appreciated the fact of how the application created visibility for queer people, and that the authentic queer stories were essential for the relevance of the concept.

For some participants the digital layer added a sense of community that was previously not there. For example, one participant explained:

"It's really strange; yes, I think I actually did [get a new understanding of the place], especially there at 'the techno night' [name of the story] because then I started to look at the place as if, well, there may have been a larger [LGBTQ+] environment for this than I thought, [I] never ever thought there should have been a queer story here, but of course, there could very well have been. I can't quite pinpoint exactly how it changed my perspective. It was a realization of standing there in that moment and feeling like, now it looks different to me." (Noah)

To Noah, the impact of being at the location associated with the story was important in order to realize that their sense of place suddenly changed. However, they found it difficult to pinpoint the exact reason. It was in the moment of being there, that Noah also realised that this indeed occurred, and as such the story added a new social meaning to the place for them and exemplifying the importance of being present at the story's location.

Another effect the stories and experience contributed to was a sense of not feeling alone; Sabrina described it as a feeling of connectedness to the community:

"I think it has potential to [make one] feel more connected; one can feel lonely among other people all the time, but that feeling of being in contact with more stories, to discover, and to see oneself in" (Sabrina)

Relatedly, Noah also noted that seeing all the pins on the map, which indicated the stories, provided them with a feeling of not being alone and fostered a connection to the greater LGBTQ+ community.

"It gives me a bit of a feeling, I know it sounds strange because I'm well aware that there are a lot of queer people in Copenhagen, but you don't see that in everyday life. So it gives a feeling of not being alone." (Noah)

Both recognized the potential value of the content being personal queer stories, as it became more relatable and created a deeper connection to the content.

Andy noted that the application prompted him to reflect upon the things he might not see; in particular, the map "create[d] a sense of community that we are all here, and we live together". Despite participant pre-existing awareness of the LGBTQ+ community, the map of stories in the application served as a reminder and representation of people they might not encounter in their daily lives. Noah also characterised it as an "invisible space" supporting awareness of the queer community existing everywhere.

4.2 Finding a Sense of Place

The participant's curiosity about the locations not only highlights the impact of being physically present at the place that the story is associated with, it also encouraged a new way to interact with the current environment. After reading the stories, Sabrina for example expressed an interest in exploring and reflecting upon the locations that the stories were associated with. She expressed how she might walk around to assign meaning to the place concerning the story she just read, and as such, the story would contribute to her experience and interaction of the place, explaining "...in this way, both the interaction with the place and the story are more intense."

Andy talked about how interacting with the story by being present at the story's location provided him with both a new understanding of the place and a connection to the anonymous author because it awakened a feeling in him that seemed familiar:

"There is a presence it evokes; suddenly, you establish a connection to this place, and to this anonymous writer. It creates an interest because one can [identify oneself with] others." (Andy)

Of the six participants, Jonathan had a more hesitant perspective, questioning how much the stories would affect his sense of a place unless the story was very significant and relatable:

"Yeah, but I don't know if it would impact my understanding, I would just recall that anecdote, but potentially if it were other stories like, if it was, 'oh I met this guy ... we had a very nice romantic story', maybe that would change it ... then I think there is potential for it to impact my opinion on that space." (Jonathan)

The participants reflected upon their experiences of reading the stories at the location the story was associated with. Their experiences illustrated how the application created a new connection to the story's place: it not only provided a feeling of presence related to the story's author and the location, but also fostered curiosity towards the physical surroundings, prompting the users to explore the place or area, and intensifying the experience of the location.

4.3 Future Connections

The overall concept of the application depended on users' level of interaction and their willingness to disclose memories, anecdotes, or other stories which are tied to specific locations. We therefore also asked participants to what extent they could imagine contributing their own stories to the application. We also left it open if they wanted to test the app by leaving a real story from their own lives. Two of the participants (Simon and Jonathan) chose to share their own personal anecdotes. Simon argued that stories could play a role in improving the community: "... there are many things to improve within our community, and stories can be helpful for that.". Similarly, Jonathan added that this could be an opportunity to release some thoughts or start a conversation: "Sometimes it really helps to verbalise or write something on paper or phone to release something or start a conversation about something." The act of sharing stories was for them a way to express their feelings towards something they might not be able to share otherwise.

Even if participants chose not to leave a story of their own during our study, they discussed how it could be potentially beneficial. To Noah, for example, the idea of adding stories suggested a certain feeling that they could contribute to others' sense of belonging: "I think I would get a feeling that I can also pass something on to others. That they also would feel comfort in it. But I think also I have a lot of stories, which I don't tell others about."

Andy mentioned that he thought the app could have been meaningful to him before he was open about his sexuality:

"There is something comforting in it, and much about the sense of community, being able to reflect oneself in others and live a bit through others if you're not quite there yet... I definitely think I missed this app, or something similar, when I was in the closet." (Andy)

Participants expressed that the application not only facilitated visibility of queer stories but also that it could serve as a platform for expressing experiences that otherwise could be challenging to communicate. By providing a digital space to share these stories, the application could serve as an outlet to express these experiences or emotions. In addition to its role of storytelling, it was also evident that the platform could play a role in offering support to those uncertain about their sexuality or gender identity.

In general, all participants expressed a strong interest in the idea of mapping queer stories. Their interview responses illustrated the impact of the application on their emotions, specifically regarding their sense of the LGBTQ+ community, sense of belonging in public spaces, and the sense of presence of the story authors.

5 DISCUSSION

One of the aims of this study was to explore how location-based queer stories could influence participants' perceptions of specific locations. Through our analysis we illustrated how it provided some of the participants with an emotion of not feeling alone, a sense of belonging, and a sense of the presence of other queer people. Because the intended users of the application were LGBTQ+ people, the stories uploaded to the application became a symbol of where these others had been and how frequented these locations were by this group. Previous research also points out how such visibility creates more trust and a sense of comfort in these locations [9],

which was also explicitly highlighted by Noah: "It gives some sort of sense of belonging, where one can relax a bit more...". Noah considered it crucial that these stories represented queer experiences in public spaces not usually identified as queer. As someone's sexuality is not always visible by the person's appearance, specific types of places have become important for LGBTQ+ people in order to find one another, develop relationships, and build community [11]. This is also reflected in how cities have formed "queer spaces" such as gay neighbourhoods, LGBTQ community houses, and queer bars [10]. It was therefore essential for the success of cultivating such places—for making the space into place of significance—that the stories presented in the app were authored by queer people. Seeing the stories on the map gave participants a sense of belonging in a public setting which they would not typically feel. The location-based aspect of the stories enabled participants to perceive locations differently and created a new sense of comfort for them.

Ulrich pointed out how the platform had the potential to show "where do queer people feel safe, and where do they not feel safe.". To Ulrich, the location-based nature of the stories could not only indicate what places are frequented by LGBTQ+ people, but also impact his understanding of what places to him could be read as safe or not. If a story described a harmful experience, they might avoid that specific space in the future.

One important effect of the app was how it added a layer of what one participant termed "invisible space" to the city. This idea relates to Duncan's writings about the heteronormativity and privatisation of sexualities: "It is usually assumed that sexuality is (and should be) confined to private spaces. This is based on the naturalization of heterosexual norms. Naturalized heterosexuality makes sexuality in public spaces nearly invisible to the straight population" [8, p. 137]. Duncan argues that public spaces are experienced in a way that obviates sexuality because of the naturalisation of heterosexuality. Our application explicitly attempts to disrupt this heteronormativity by placing queer experiences in a public (virtual) space. To participants, it was comforting that what would normally be invisible to them suddenly became visible through the application.

Although it could be argued that the application does not influence public spaces because it only exists within virtual space, contemporary accounts of public spaces should consider the idea of net locality: the notion that it is no longer possible to address distinctions between physical and digital spaces because digital information becomes an essential part of people's experiences of urban spaces [9]. As such, the digital association of queer stories with physical locations becomes an important part of the experience of the physical space for those with access to this information. This was also reflected in the findings that the application assigned a new layer of meaning to the places the participants visited; whether digital or not, it still contributed to the understanding of the place in question. Thus our application contributed towards transforming mere physical spaces into meaningful and culturally important places.

6 CONCLUSION

The purpose of this study was to explore how to facilitate more visibility of the LGBTQ+ community by using mobile technologies

and engaging users with location-based queer stories. The findings illustrated how our application, Queer Annotations, fostered a sense of belonging within the community, reducing the feeling of being alone in public spaces, and working as a reminder of other queer people's presence.

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